

LAPORAN ULANGAN AKHIR SEMESTER
BASIS DATA II

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2026

Deskripsi Kasus :

Lembaga Sertifikasi Profesi (LSP) akan membuat sistem informasi untuk mengelola proses sertifikasi kompetensi. Dalam sistem tersebut terdapat data:

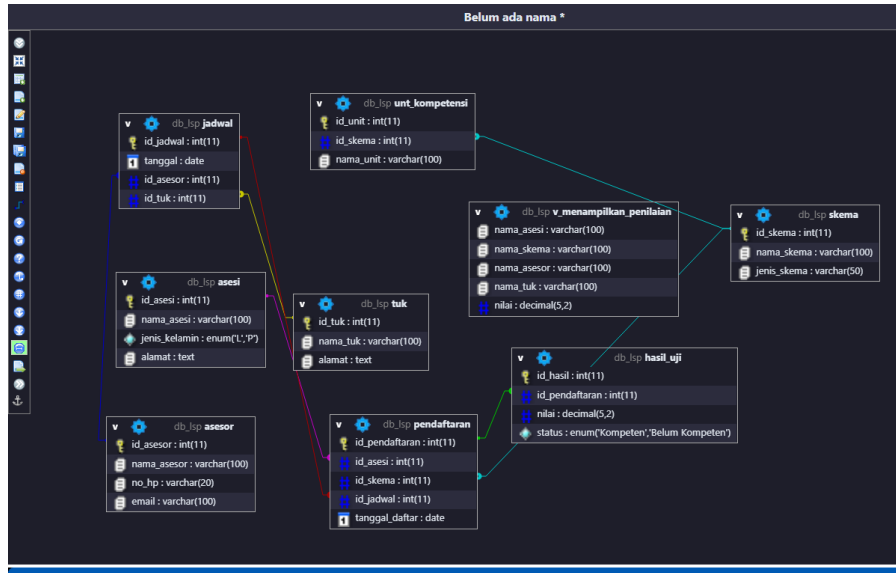
Asesor, Asesi (Peserta), Skema Sertifikasi, Unit Kompetensi, Jadwal Uji Kompetensi, Tempat Uji Kompetensi (TUK), Pendaftaran Sertifikasi, Hasil Uji Kompetensi

Satu skema memiliki banyak unit kompetensi. Seorang asesi dapat mengikuti lebih dari satu skema. Setiap jadwal memiliki satu asesor dan satu TUK.

1. Buatlah Entity Relationship Diagram (ERD) minimal terdiri dari tabel berikut. asesor, asesi, skema, unit_kompetensi, tuk, jadwal, pendaftaran, hasil_uji, Lengkapi dengan, Primary Key, Foreign Key (bobot (20%).
2. Implementasikan database dengan nama : db_lsp(bobot (20%).
3. Buat seluruh tabel menggunakan Command-Line Interface (CLI) SQL.(bobot (20%).
4. Masukkan data minimal 5 asesor, 20 asesi, 5 skema, 20 unit kompetensi, 5 TUK, 10 jadwal, 20 pendaftaran, 20 hasil_uji INSERT INTO (bobot (20%).
5. Buat sebuah VIEW bernama Menampilkan Nama Asesi, Nama Skema, Nama Asesor, Nama TUK, Nilai (bobot (10%).
6. Gunakan Subquery Menampilkan peserta dengan nilai di atas rata-rata. Menampilkan asesor yang belum memiliki jadwal. Menampilkan skema dengan peserta terbanyak.(bobot (10%).)

Jawaban :

1. Entity Realtionship Diagram.



2. Implementasi Database

```
MariaDB [(none)]> CREATE DATABASE db_lsp;  
Query OK, 1 row affected (0.004 sec)
```

3. Membuat Tabel

a. Tabel Asesor

```
MariaDB [db_lsp]> CREATE TABLE asesor (  
  -> id_asesor INT AUTO_INCREMENT PRIMARY KEY,  
  -> nama_asesor VARCHAR(100),  
  -> no_hp VARCHAR(20),  
  -> email VARCHAR(100)  
  -> );  
Query OK, 0 rows affected (0.044 sec)
```

b. Tabel Asesi

```
MariaDB [db_lsp]> CREATE TABLE asesi (  
  -> id_asesi INT AUTO_INCREMENT PRIMARY KEY,  
  -> nama_asesi VARCHAR(100),  
  -> jenis_kelamin ENUM('L','P'),  
  -> alamat TEXT  
  -> );  
Query OK, 0 rows affected (0.029 sec)
```

c. Tabel Skema

```
MariaDB [db_lsp]> INSERT INTO skema (nama_skema,jenis_skema) VALUES
-> ('Junior Web Developer','BNSP'),
-> ('Junior Network Administrator','BNSP'),
-> ('Database Programmer','BNSP'),
-> ('Cyber Security','BNSP'),
-> ('Digital Marketing','BNSP');
Query OK, 5 rows affected (0.005 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

d. Tabel TUK

```
MariaDB [db_lsp]> CREATE TABLE tuk (
-> id_tuk INT AUTO_INCREMENT PRIMARY KEY,
-> nama_tuk VARCHAR(100),
-> alamat TEXT
-> );
Query OK, 0 rows affected (0.042 sec)
```

e. Tabel Jadwal

```
MariaDB [db_lsp]> CREATE TABLE jadwal (
-> id_jadwal INT AUTO_INCREMENT PRIMARY KEY,
-> tanggal DATE,
-> id_asesor INT,
-> id_tuk INT,
-> FOREIGN KEY(id_asesor) REFERENCES asesor(id_asesor),
-> FOREIGN KEY(id_tuk) REFERENCES tuk(id_tuk)
-> );
Query OK, 0 rows affected (0.040 sec)
```

f. Tabel Pendaftaran

```
MariaDB [db_lsp]> CREATE TABLE pendaftaran (
-> id_pendaftaran INT AUTO_INCREMENT PRIMARY KEY,
-> id_asesi INT,
-> id_skema INT,
-> id_jadwal INT,
-> tanggal_daftar DATE,
-> FOREIGN KEY(id_asesi) REFERENCES asesori(id_asesi),
-> FOREIGN KEY(id_skema) REFERENCES skema(id_skema),
-> FOREIGN KEY(id_jadwal) REFERENCES jadwal(id_jadwal)
-> );
Query OK, 0 rows affected (0.038 sec)
```

g. Tabel Unit Kompetensi

```
MariaDB [db_lsp]> CREATE TABLE unt_kompetensi (
-> id_unit INT AUTO_INCREMENT PRIMARY KEY,
-> id_skema INT,
-> nama_unit VARCHAR(100),
-> FOREIGN KEY(id_skema) REFERENCES skema(id_skema)
-> );
Query OK, 0 rows affected (0.205 sec)
```

h. Tabel Hasil Uji

```
MariaDB [db_lsp]> CREATE TABLE hasil_uji(  
  -> id_hasil INT AUTO_INCREMENT PRIMARY KEY,  
  -> id_pendaftaran INT,  
  -> nilai DECIMAL(5,2),  
  -> status ENUM('Kompeten','Belum Kompeten'),  
  -> FOREIGN KEY (id_pendaftaran) REFERENCES pendaftaran(id_pendaftaran)  
  -> );  
Query OK, 0 rows affected (0.038 sec)
```

4. Insert Data

```
MariaDB [db_lsp]> INSERT INTO asesor(nama_asesor,no_hp,email) VALUES  
  -> ('Andi','081111111','andi@gmail.com'),  
  -> ('Budi','082222222','budi@gmail.com'),  
  -> ('Candra','083333333','candra@gmail.com'),  
  -> ('Deni','084444444','deni@gmail.com'),  
  -> ('Eko','085555555','eko@gmail.com');  
Query OK, 5 rows affected (0.027 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_lsp]> INSERT INTO asesi (nama_asesi,jenis_kelamin,alamat) VALUES  
  -> ('Ahmad Fauzi','L','Tegal'),  
  -> ('Rina Putri','P','Brebes'),  
  -> ('Yusuf Hidayat','L','Pemalang'),  
  -> ('Sinta Dewi','P','Slawi'),  
  -> ('Doni Setiawan','L','Tegal'),  
  -> ('Nabila Zahra','P','Brebes'),  
  -> ('Fajar Nugroho','L','Pemalang'),  
  -> ('Lilis Handayani','P','Slawi'),  
  -> ('Rizky Maulana','L','Tegal'),  
  -> ('Fitri Anisa','P','Brebes'),  
  -> ('Arif Rahman','L','Pemalang'),  
  -> ('Maya Sari','P','Slawi'),  
  -> ('Galih Pratama','L','Tegal'),  
  -> ('Nur Aisyah','P','Brebes'),  
  -> ('Ilham Ramadhan','L','Pemalang'),  
  -> ('Dinda Maharani','P','Slawi'),  
  -> ('Bagus Saputra','L','Tegal'),  
  -> ('Intan Permata','P','Brebes'),  
  -> ('Wahyu Kurniawan','L','Pemalang'),  
  -> ('Putri Ayu','P','Slawi');  
Query OK, 20 rows affected (0.006 sec)  
Records: 20 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_lsp]> INSERT INTO skema (nama_skema,jenis_skema) VALUES  
  -> ('Junior Web Developer','BNSP'),  
  -> ('Junior Network Administrator','BNSP'),  
  -> ('Database Programmer','BNSP'),  
  -> ('Cyber Security','BNSP'),  
  -> ('Digital Marketing','BNSP');  
Query OK, 5 rows affected (0.005 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

```

MariaDB [db_lsp]> INSERT INTO jadwal (tanggal,id_asesor,id_tuk) VALUES
-> ('2026-08-01',1,1),
-> ('2026-08-02',2,2),
-> ('2026-08-03',3,3),
-> ('2026-08-04',4,4),
-> ('2026-08-05',5,5),
-> ('2026-08-06',1,2),
-> ('2026-08-07',2,3),
-> ('2026-08-08',3,4),
-> ('2026-08-09',4,5),
-> ('2026-08-10',5,1);
Query OK, 10 rows affected (0.019 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```

MariaDB [db_lsp]> INSERT INTO tuk (nama_tuk,alamat) VALUES
-> ('TUK Purbaya','Slawi'),
-> ('TUK Tegal','Tegal'),
-> ('TUK Brebes','Brebes'),
-> ('TUK Pemalang','Pemalang'),
-> ('TUK Pekalongan','Pekalongan');
Query OK, 5 rows affected (0.008 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

```

MariaDB [db_lsp]> INSERT INTO unt_kompetensi (id_skema,nama_unit) VALUES
-> (1,'HTML'),
-> (1,'CSS'),
-> (1,'JavaScript'),
-> (1,'PHP'),
-> (2,'Dasar Jaringan'),
-> (2,'Routing'),
-> (2,'Switching'),
-> (2,'Subnetting'),
-> (3,'MySQL'),
-> (3,'SQL Query'),
-> (3,'Stored Procedure'),
-> (3,'Database Design'),
-> (4,'Linux Security'),
-> (4,'Firewall'),
-> (4,'Penetration Testing'),
-> (4,'Network Security'),
-> (5,'SEO'),
-> (5,'Google Ads'),
-> (5,'Content Marketing'),
-> (5,'Social Media Marketing');
Query OK, 20 rows affected (0.024 sec)
Records: 20 Duplicates: 0 Warnings: 0

```

```

MariaDB [db_lsp]> INSERT INTO pendaftaran
-> (id_asesi,id_skema,id_jadwal,tanggal_daftar)
-> VALUES
-> (1,1,1,'2026-07-20'),
-> (2,2,2,'2026-07-20'),
-> (3,3,3,'2026-07-20'),
-> (4,4,4,'2026-07-20'),
-> (5,5,5,'2026-07-20'),
-> (6,1,6,'2026-07-21'),
-> (7,2,7,'2026-07-21'),
-> (8,3,8,'2026-07-21'),
-> (9,4,9,'2026-07-21'),
-> (10,5,10,'2026-07-21'),
-> (11,1,1,'2026-07-22'),
-> (12,2,2,'2026-07-22'),
-> (13,3,3,'2026-07-22'),
-> (14,4,4,'2026-07-22'),
-> (15,5,5,'2026-07-22'),
-> (16,1,6,'2026-07-23'),
-> (17,2,7,'2026-07-23'),
-> (18,3,8,'2026-07-23'),
-> (19,4,9,'2026-07-23'),
-> (20,5,10,'2026-07-23');
Query OK, 20 rows affected (0.017 sec)
Records: 20  Duplicates: 0  Warnings: 0

```

```

MariaDB [db_lsp]> INSERT INTO hasil_uji
-> (id_pendaftaran,nilai,status)
-> VALUES
-> (1,85,'Kompeten'),
-> (2,78,'Kompeten'),
-> (3,90,'Kompeten'),
-> (4,70,'Belum Kompeten'),
-> (5,88,'Kompeten'),
-> (6,92,'Kompeten'),
-> (7,67,'Belum Kompeten'),
-> (8,81,'Kompeten'),
-> (9,76,'Kompeten'),
-> (10,95,'Kompeten'),
-> (11,73,'Belum Kompeten'),
-> (12,84,'Kompeten'),
-> (13,89,'Kompeten'),
-> (14,69,'Belum Kompeten'),
-> (15,91,'Kompeten'),
-> (16,87,'Kompeten'),
-> (17,75,'Kompeten'),
-> (18,82,'Kompeten'),
-> (19,64,'Belum Kompeten'),
-> (20,93,'Kompeten');
Query OK, 20 rows affected (0.018 sec)
Records: 20  Duplicates: 0  Warnings: 0

```


5. Tabel View

```
MariaDB [db_lsp]> CREATE VIEW v_menampilkan_penilaian AS
-> SELECT
-> a.nama_asesi,
-> s.nama_skema,
-> asr.nama_asesor,
-> t.nama_tuk,
-> h.nilai
-> FROM hasil_uji h
-> JOIN pendaftaran p
-> ON h.id_pendaftaran = p.id_pendaftaran
-> JOIN asesi a ON p.id_asesi = a.id_asesi
-> JOIN skema s ON p.id_skema = s.id_skema
-> JOIN jadwal j ON p.id_jadwal = j.id_jadwal
-> JOIN asesor asr ON j.id_asesor = asr.id_asesor
-> JOIN tuk t ON j.id_tuk = t.id_tuk;
Query OK, 0 rows affected (0.066 sec)
```

```
MariaDB [db_lsp]> SELECT * FROM v_menampilkan_penilaian;
```

nama_asesi	nama_skema	nama_asesor	nama_tuk	nilai
Ahmad Fauzi	Junior Web Developer	Andi	TUK Purbaya	85.00
Nabila Zahra	Junior Web Developer	Andi	TUK Tegal	92.00
Arif Rahman	Junior Web Developer	Andi	TUK Purbaya	73.00
Dinda Maharani	Junior Web Developer	Andi	TUK Tegal	87.00
Rina Putri	Junior Network Administrator	Budi	TUK Tegal	78.00
Fajar Nugroho	Junior Network Administrator	Budi	TUK Brebes	67.00
Maya Sari	Junior Network Administrator	Budi	TUK Tegal	84.00
Bagus Saputra	Junior Network Administrator	Budi	TUK Brebes	75.00
Yusuf Hidayat	Database Programmer	Candra	TUK Brebes	90.00
Lilis Handayani	Database Programmer	Candra	TUK Pemalang	81.00
Galih Pratama	Database Programmer	Candra	TUK Brebes	89.00
Intan Permata	Database Programmer	Candra	TUK Pemalang	82.00
Sinta Dewi	Cyber Security	Deni	TUK Pemalang	70.00
Rizky Maulana	Cyber Security	Deni	TUK Pekalongan	76.00
Nur Aisyah	Cyber Security	Deni	TUK Pemalang	69.00
Wahyu Kurniawan	Cyber Security	Deni	TUK Pekalongan	64.00
Doni Setiawan	Digital Marketing	Eko	TUK Pekalongan	88.00
Fitri Anisa	Digital Marketing	Eko	TUK Purbaya	95.00
Ilham Ramadhan	Digital Marketing	Eko	TUK Pekalongan	91.00
Putri Ayu	Digital Marketing	Eko	TUK Purbaya	93.00

20 rows in set (0.013 sec)

6. Perintah SubQuery

a. Subquery Dengan Nilai Diatas Rata Rata

```
MariaDB [db_lsp]> SELECT
-> a.nama_asesi,
-> h.nilai
-> FROM asesori a
-> JOIN pendaftaran p ON a.id_asesi = p.id_asesi
-> JOIN hasil_uji h ON p.id_pendaftaran = h.id_pendaftaran
-> WHERE nilai >
-> (
-> SELECT AVG(nilai) FROM hasil_uji
-> );
```

nama_asesi	nilai
Ahmad Fauzi	85.00
Yusuf Hidayat	90.00
Doni Setiawan	88.00
Nabila Zahra	92.00
Fitri Anisa	95.00
Maya Sari	84.00
Galih Pratama	89.00
Ilham Ramadhan	91.00
Dinda Maharani	87.00
Intan Permata	82.00
Putri Ayu	93.00

11 rows in set (0.008 sec)

b. Subquery Menampilkan Asesor Yang belum Memiliki Jadwal

```
MariaDB [db_lsp]> SELECT * FROM asesori
-> WHERE id_asesor NOT IN (
-> SELECT id_asesor FROM jadwal
-> );
```

Empty set (0.005 sec)

c. Subquery Menampilkan Skema Dengan Peserta Terbanyak

```
MariaDB [db_lsp]> SELECT nama_skema FROM skema
-> WHERE id_skema = (
-> SELECT id_skema
-> FROM pendaftaran
-> GROUP BY id_skema
-> ORDER BY COUNT(*) DESC
-> LIMIT 1
-> );
```

nama_skema
Junior Web Developer

1 row in set (0.003 sec)

```
MariaDB [db_lsp]>
```